Lab - Research a Hardware Upgrade

Introduction

Use the Internet, a newspaper, or a local store to gather information about hardware components. Your customer's computer currently has one module of 2 GB of RAM, a 500 GB hard disk drive, and a PCIe video adapter card with 256 MB of RAM. Your customer wants to be able to play advanced video games.

Instructions

Step 1: Research memory options.

Shop around, and in the table below list the brand, model number, features, and cost for two different 8 GB modules of DDR3.

Brand and Model Number	Features	Cost
Kingston KVR16N11/8	*JEDEC standard 1.5V Power Supply, VDDQ = 1.5 V, 800 MHz fCK for 1600Mb/sec/pin*	*1,000 - 1,350 PHP*
Team Elite TED38G1600C11BK	*Voltage = 1.5 V, Speed = 1600 MHz, CAS latency = C 11-11-11-28, 8GB capacity*	*2,000 - 2,500 PHP*

Question:

Based on your research, which RAM would you select? Be prepared to discuss your decisions regarding the RAM you select.

- Based on my research, I would personally choose the Kingston KVR16N11/8 because if we compare both of their features, we can see that they are very similar in most of the specifications mentioned, but Kingston has an edge with its lower price range. Also, I have found that Kingston is a very common and trusted brand and is widely available here in the Philippines.

Step 2: Research hard disk drive options.

Shop around, and in the table below list the brand, model number, features, and cost for a 3 TB 7200 rpm SATA 3 hard disk drive and a 500GB or 1TB solid state drive (SSD).

Brand and Model Number	Features	Cost
Seagate BarraCuda ST3000DM007 HDD	*Size: 3TB, Internal Data Rate: 190 MBps, Drive Transfer Rate: 600 MBps, Average Latency: 6ms*	*5,000 - 5,400 PHP*
Kingston NV3 1TB NVMe SSD	*Size: 1TB, Endurance (Total Bytes Written): 320TB, Form Factor: M.2 2280, Interface: PCIe 4.0 x4 NVMe, Speed (Sequential read/write): 6,000/4,000MB/s*	*10,000 - 12,000 PHP*

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Question:

Based on your research, which hard disk drive would you select? Be prepared to discuss your decisions regarding the hard disk drive you select.

- Based on my research, I would choose the Kingston NV3 1TB NVMe SSD as my research suggested that SSDs are generally faster than HDDs. The Kingston SSD provides much faster boot times, game loading, and overall system performance than a hard disk drive.

Step 3: Research video adapter card options.

Shop around, and in the table below list the brand, model number, features, and cost for two different PCIe video adapter cards with at least 1 GB of RAM.

Brand and Model Number	Features	Cost
Gigabyte NVIDIAÂ® GeForce GTX 1650 D6 OC GV-N1656OC-4GD	*Video Memory: 4GB, GDDR6, Clock speed: 1635 MHz, Bus Interface: PCIe 3.0 x16*	*9,150 PHP*
PowerColor Radeon RX 6400 AXRX 6400 4GBD6-DH 64BIT GDDR6 Gaming Videocard	*Video Memory: 4GB, Clock speed: up to 2049 MHz, Bus Interface: PCIE 4.0*	*5,599 PHP*

Question:

Based on your research, which video adapter card would you select? Be prepared to discuss your decisions regarding the video adapter card you select.

- Based on my research, I would choose the Gigabyte NVIDIAÂ® GeForce GTX 1650 D6 OC GV-N1656OC-4GD, as it suggests better gaming performance. It is also an overclocked version so it slightly provides higher performance.

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